

DESIGNER'S NOTES

Glimmerdark is a 2–4 player game for one standard deck, a board and some tokens. This is how it was balanced: a rules-exact simulator played the game to a pulp, and every change to the rules had to earn its place with data. The rules went through four versions; the table compares the first draft with the final rules (v4), and the rest of these notes show the working.

	v1: first draft	v4: final rules	Target
Median playtime, 2 / 3 / 4 players	48 / 41 / 38 min	33 / 36 / 35 min	30–45 min
Games inside 30–45 min	35% / 74% / 99%	81% / 87% / 100%	most
Furthest delver from a fair share	29.0 / 20.5 / 15.3 pts	4.1 / 1.8 / 2.3 pts	≤ 5 pts
First player's wins ÷ fair share	1.06 / 1.10 / 1.10	1.00 / 1.00 / 0.99	≈ 1.00
Halfway leader goes on to win	79% / 65% / 55%	69% / 56% / 48%	lower
Planning AI vs cautious AI (2p)	48%	62%	> 50%
Greedy AI vs cautious AI (2p)	32%	52%	near 50%

Delver balance is each character's win rate minus a fair share (50% / 33.3% / 25%), in percentage points. Every figure comes from 10,000 games per line-up.

HOW THE GAME WAS TESTED

The simulator (sim/glimmer/) enforces every rule in the rulebook: a real 52-card deck plus two jokers, dealt, drawn, discarded and reshuffled; the Warden's rumble, targeting, route and crush; every delver ability; tremors, surges and the collapse. An illegal play raises an error instead of being quietly allowed, and a test suite checks the deck, the board and each rule in isolation.

Four AI players. *Greedy* grabs the most glimmer it can and takes risks. *Cautious* banks early and keeps away from the Warden. *Tactician* plans each turn around its delver's ability, the Warden's likely next move and who is winning. *Random* plays any legal card and is used only to measure luck. Balance numbers use Tactician in every seat, since it plays each ability the way it is meant to be played.

The sweep. For each version: every line-up of 2, 3 and 4 delvers from the six (50 line-ups) at 10,000 games each, with seats shuffled; 13 strategy matchups and 3 luck tests at 10,000 games each; a separate 10,000-games-per-player-count playtime run; and each ability duelled against a delver with no ability. In all, 3,000,000 simulated games across the four versions. At that size, a delver's win rate for one player count is within about ±0.5 points (95% confidence).

Playtime model. The simulator counts turns, not minutes. A player turn (play up to two cards, resolve them, draw back to five) is costed at 45 seconds, with 30 s for a table that knows the game and 60 s for a slow or new one. Each Warden phase (flip a card, walk the Warden, resolve a crush) adds 10 s. Setup and teaching are not included. The minutes in these notes are that formula applied to the simulated turn counts.

THE BALANCING STORY

Each entry shows what the previous version's full sweep said, what changed because of it, and what the next sweep showed. The same log is in `CHANGELOG.md`.

V1: FIRST DRAFT, EXACTLY AS WRITTEN

The rules in `design/rules_v1_first_draft.md`, written before any simulation.

Result: the v1 sweep

- Playtime at 45 s a turn: median 48 / 41 / 38 min (2 / 3 / 4 players); 35% / 74% / 99% of games inside 30–45 min.
- Delver balance: best-to-worst spread 47.8 / 32.7 / 24.9 points; furthest from a fair share 29.0 / 20.5 / 15.3 points.
- Outside ± 5 points: Mira -29.0 (2p), Gritch +18.8 (2p), Sable +7.9 (2p), Mira -20.5 (3p), Gritch +12.2 (3p), Sable +7.2 (3p), Old Quill +6.7 (3p), Mira -15.3 (4p), Old Quill +9.5 (4p), Gritch +7.8 (4p).
- First player wins 52.8% / 36.8% / 27.5% (fair 50% / 33.3% / 25%).
- Halfway leader wins 79% / 65% / 55%; lead changes per game 2.8 / 3.2 / 2.9.
- Planning AI wins 68% against greedy and 48% against cautious; greedy wins 32% against cautious (2 players).
- Crushes per game 11.0 / 11.8 / 11.7.

V2: SHORTER, LESS SNOWBALLY, AND A REASON TO GO DEEP

Change	Why (data from v1)
The collapse track: jokers now move the marker too (a Tremor), and the end space depends on player count (7 / 8 / 9 with 2 / 3 / 4 players) instead of 3 reshuffles for everyone.	Median playtime was 48 / 41 / 38 minutes with 2 / 3 / 4 players, and only 35% of 2-player games fitted the 30–45 minute window.
The Warden walks toward the delver who is winning (vault plus pack), not the one carrying the most.	The player leading at the halfway point won 79% / 65% / 55% of games. The Warden is the game's catch-up mechanism, so it should lean on the leader.
Deep glimmer is worth 4, not 3.	Going deep didn't pay: the cautious AI beat the greedy one 68% of the time and beat the planning AI 52%. Sitting in the shallows was the best strategy, which is dull.
The first player's first turn has 1 action instead of 2.	The first player won 52.8% of 2-player games (fair: 50%).
Mira: the first ♥ she plays to Mine each turn costs no action (hearts still match every vein).	Mira was -29.0 / -20.5 / -15.3 points from a fair share (2 / 3 / 4 players). Duelling a delver with no ability at all, Mira won 47.1% of games: letting hearts match any vein was no better than having no ability, because the scarce thing on a turn is movement, not matching.
Gritch: a ♠ Move takes him up to 2 chambers instead of 3.	Gritch was +18.8 / +12.2 / +7.8 points from a fair share: in a game where movement is the bottleneck, triple moves were too much.

Result: the v2 sweep

- Playtime at 45 s a turn: median 33 / 36 / 35 min (2 / 3 / 4 players); 86% / 88% / 100% of games inside 30–45 min.
- Delver balance: best-to-worst spread 9.5 / 12.8 / 12.3 points; furthest from a fair share 5.3 / 7.1 / 6.7 points.
- Outside ± 5 points: Mira +5.3 (2p), Mira +7.1 (3p), Pip & Pell -5.8 (3p), Mira +6.7 (4p), Pip & Pell -5.7 (4p), Gritch -5.3 (4p).
- First player wins 49.1% / 33.3% / 24.9% (fair 50% / 33.3% / 25%).
- Halfway leader wins 68% / 56% / 48%; lead changes per game 2.9 / 3.4 / 3.2.
- Planning AI wins 60% against greedy and 60% against cautious; greedy wins 50% against cautious (2 players).
- Crushes per game 8.9 / 10.8 / 12.1.

V3: RETUNING THE DELVERS

Change	Why (data from v2)
Gritch: back to 3 chambers per ♠, but he can't pass through the Gate (he may stop there), and his pack holds 4 instead of 5.	At 2 chambers Gritch was -1.1 / -4.8 / -5.3. Speed is his identity, so it came back, and the cost moved to how much he can haul and how easily he banks.
Hulda: a high ♣ (7–10, or the Jack, which matches as a 10) played as Pilfer takes 2 glimmer instead of 1.	Hulda was -2.7 / -4.0 / -3.6.
Pip & Pell: twice per turn instead of once, and a suited run (same suit, next rank up or down) counts as well as a pair.	Pip & Pell were -4.2 / -5.8 / -5.7; pairs in a five-card hand are too rare to lean on.
Old Quill: his Jack Shortcut moves 4 chambers instead of 3.	Old Quill was -1.4 / +2.8 / +4.7. Win rates are shared, so raising three delvers pulls the other three down; Quill was already below par with 2 players and would have sunk further, so he got a small boost too.
No change to Mira or Sable.	They led: Mira +5.3 / +7.1 / +6.7, Sable +4.0 / +4.8 / +3.2. Rather than weaken abilities that play well, the trailing delvers were brought up to meet them.
The first player's short first turn now applies only with 3 or 4 players.	The first player won 49.1% / 33.3% / 24.9% with 2 / 3 / 4 players (fair: 50% / 33.3% / 25%). With 2 players the one-action first turn over-corrected; with 3 and 4 it was spot on.

Result: the v3 sweep

- Playtime at 45 s a turn: median 33 / 36 / 35 min (2 / 3 / 4 players); 81% / 87% / 100% of games inside 30–45 min.
- Delver balance: best-to-worst spread 8.0 / 2.0 / 3.4 points; furthest from a fair share 4.5 / 1.2 / 1.9 points.
- Outside ±5 points: none.
- First player wins 50.1% / 33.4% / 24.8% (fair 50% / 33.3% / 25%).
- Halfway leader wins 69% / 56% / 48%; lead changes per game 2.8 / 3.3 / 3.1.
- Planning AI wins 61% against greedy and 62% against cautious; greedy wins 52% against cautious (2 players).
- Crushes per game 8.8 / 10.6 / 11.9.

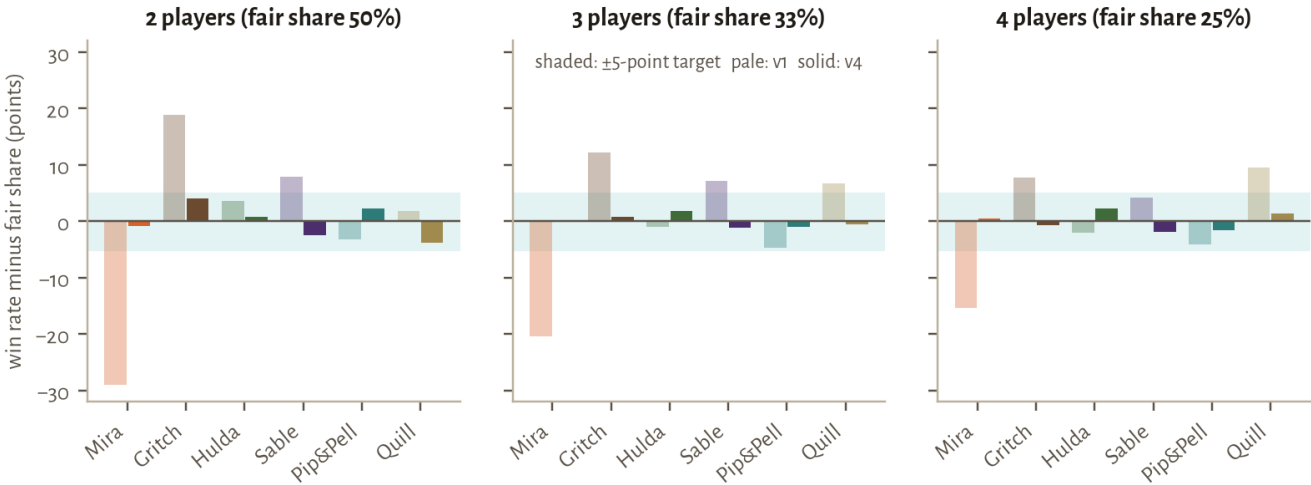
V4: ONE CLEAN RULE FOR HULDA

Change	Why (data from v3)
Hulda: face-card ♣ count as high too, so any ♣ of 7 or more takes 2.	In v3 the J♣ counted as high (a Jack matches as a 10) but the Q♣ and K♣ did not, which was hard to remember and harder to print. Hulda was -0.9 / -0.4 / +0.2 before the change.

Result: the v4 sweep

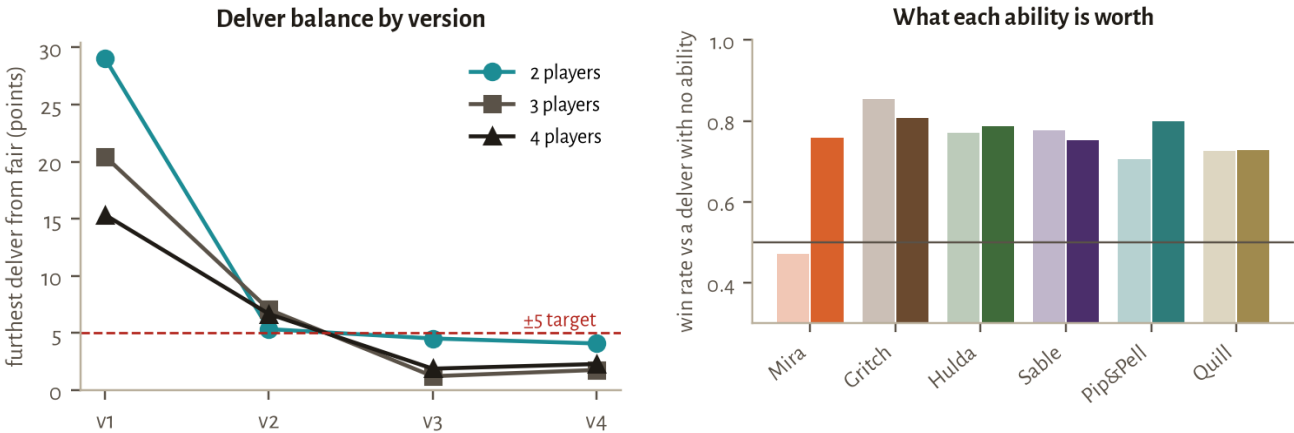
- Playtime at 45 s a turn: median 33 / 36 / 35 min (2 / 3 / 4 players); 81% / 87% / 100% of games inside 30–45 min.
- Delver balance: best-to-worst spread 7.9 / 3.0 / 4.2 points; furthest from a fair share 4.1 / 1.8 / 2.3 points.
- Outside ±5 points: none.
- First player wins 49.9% / 33.3% / 24.7% (fair 50% / 33.3% / 25%).
- Halfway leader wins 69% / 56% / 48%; lead changes per game 2.8 / 3.3 / 3.1.
- Planning AI wins 61% against greedy and 62% against cautious; greedy wins 52% against cautious (2 players).
- Crushes per game 8.8 / 10.6 / 11.9.

DELVERS BEFORE AND AFTER



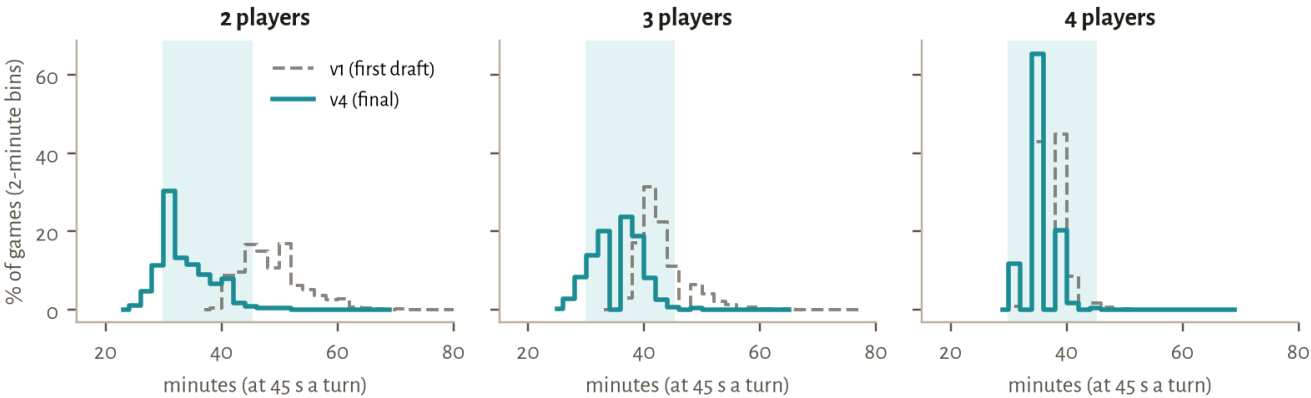
Delver	v1: 2p / 3p / 4p	v2: 2p / 3p / 4p	v3: 2p / 3p / 4p	v4: 2p / 3p / 4p
Mira	-29.0 / -20.5 / -15.3	+5.3 / +7.1 / +6.7	-0.5 / +0.6 / +1.0	-0.9 / +0.2 / +0.5
Gritch	+18.8 / +12.2 / +7.8	-1.1 / -4.8 / -5.3	+4.5 / +1.2 / -0.4	+4.1 / +0.8 / -0.7
Hulda	+3.7 / -0.9 / -2.1	-2.7 / -4.0 / -3.6	-0.9 / -0.4 / +0.2	+0.8 / +1.8 / +2.3
Sable	+7.9 / +7.2 / +4.3	+4.0 / +4.8 / +3.2	-2.3 / -0.8 / -1.5	-2.4 / -1.2 / -1.9
Pip & Pell	-3.3 / -4.7 / -4.2	-4.2 / -5.8 / -5.7	+2.7 / -0.7 / -1.2	+2.3 / -1.1 / -1.6
Old Quill	+1.9 / +6.7 / +9.5	-1.4 / +2.8 / +4.7	-3.5 / +0.0 / +1.9	-3.8 / -0.5 / +1.3

Points from a fair share for every version; red is outside the ± 5 -point target.



Right: each delver duelling an ability-less delver, 2 players, 10,000 games each. Pale bars are v1, solid bars v4. In the final rules each ability wins between 73% and 81% of these duels: all worth having, none a lock.

GAME LENGTH BEFORE AND AFTER



Share of games finishing at each minute, at 45 s a turn. Shaded: the 30–45 minute target.

Final rules (v4)

Players	Turns (median)	Rounds	Minutes at 45 s: p10 / median / p90	Median at 30 s / 60 s	Inside 30–45
2	40	20.1	28 / 33 / 40	23 / 43	81%
3	45	14.6	29 / 36 / 41	25 / 48	87%
4	44	11.1	32 / 35 / 38	24 / 46	100%

First draft (v1)

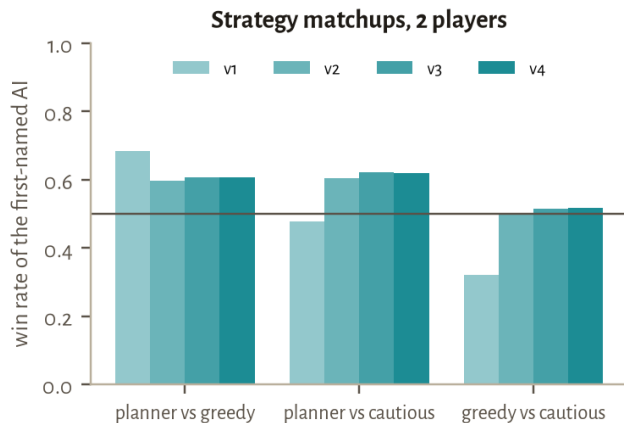
Players	Turns (median)	Rounds	Minutes at 45 s: p10 / median / p90	Median at 30 s / 60 s	Inside 30–45
2	58	29.3	43 / 48 / 57	34 / 63	35%
3	51	17.9	39 / 41 / 48	28 / 54	74%
4	48	11.7	35 / 38 / 41	26 / 50	99%

Why the collapse clock works: the marker moves on every reshuffle and every joker, so a game lasts about the same number of trips through the deck however the players behave. That is why no simulated game stalls: the longest final-rules game is set by the deck, not by anyone's caution. With more players each delver gets fewer turns per trip through the deck, so the end space grows with the player count (7 / 8 / 9) to give everyone enough turns while keeping each count in the window.

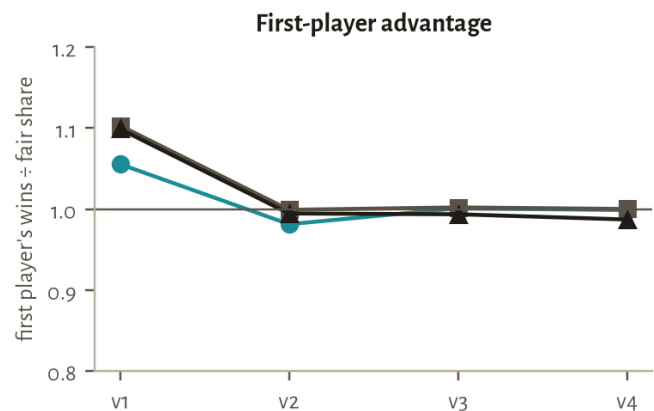
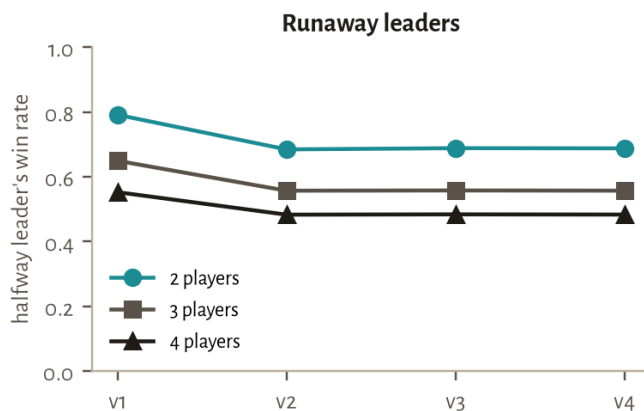
The 2-player games that miss the window mostly miss it short: 17% finish under 30 minutes and 2% run past 45 (at 45 s a turn). A slower table lands them inside it.

Games that hit the simulator's safety limit without ending: 0 (all versions).

STRATEGY, SNOWBALLING AND SEATS



No single strategy should dominate. In the first draft, turtling won: the cautious AI beat the greedy one 68% of the time and beat the planner 52%. Deeper, richer glimmer and a Warden that hunts the leader were the answer. In the final rules the planner wins 61% against greedy play and 62% against cautious play, so thinking ahead pays without guaranteeing anything. Greedy and cautious play are close (greedy wins 52%): both are viable styles, neither is a trap.



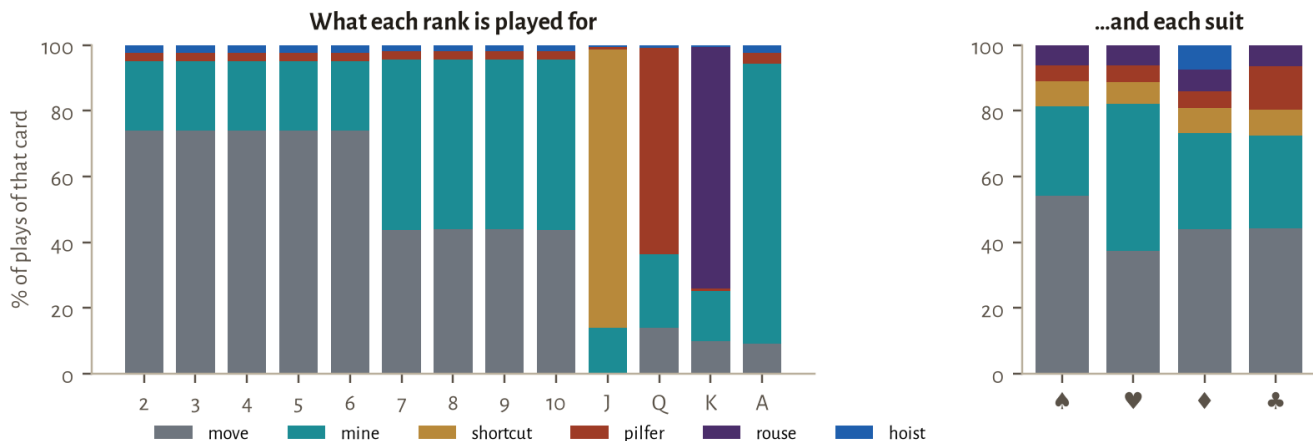
Left: how often whoever leads at the halfway round goes on to win (a pure coin-flip game would sit near the fair share; a runaway game near 100%). Right: the first player's win rate over a fair share (1.00 = no advantage).

In the final rules the halfway leader wins 69% / 56% / 48% of 2 / 3 / 4-player games, down from 79% / 65% / 55%. Being ahead is an advantage, as it should be, but the Warden's hunt and the pack-dropping crush keep leads contestable: the lead changes hands 2.8 / 3.3 / 3.1 times a game on average.

LUCK VERSUS SKILL

Two tests. First, the planning AI against an AI that plays random legal cards: the planner wins 99.9% of 2-player games and 99.9% of 4-player games against three random players, so decisions matter far more than the deal. Second, among equally skilled players, the one whose draws happened to hold the biggest share of high cards (7 and up, faces and aces included) wins 58% of 2-player games (fair: 50%), and across all balance games 57% / 40% / 31% with 2 / 3 / 4 players (fair 50% / 33% / 25%). A good draw helps; it doesn't decide the game.

CARDS: EVERY RANK HAS A JOB



How often each card was played for each action across every final-rules balance game. Low cards (2–6) are the movers (74% of their plays), high cards (7–10) the miners (52%), and each face card does its own job: Jacks shortcut (85%), Queens pilfer (63%), Kings rouse the Warden (73%). Aces match any vein, so they are mostly spent mining (85%). Every rank gets played: the least-played comes up 74% as often as an even share would give. 12 of the 13 ranks are regularly played two or more different ways (each at least 10% of the time), so a card is rarely stuck with one use; the exception: Aces, which are wild and nearly always mine. The suits look alike apart from their delver's pull: ♠ 54% moves, ♥ 45% mines, ♦ 7% hoists, ♣ 13% pilfers.

WHAT'S LEFT, AND WHY IT STAYS

In the final rules every delver is within ± 4.1 points of a fair share with 2 players, ± 1.8 with 3 and ± 2.3 with 4. The largest remaining gaps:

- **Gritch, 2 players: +4.1 points** (+0.8 with 3, -0.7 with 4). His edge is speed, and speed pays per turn: each delver gets about 20 turns in a 2-player game against 11 with 4, so a faster delver pulls further ahead over a longer personal game. Any nerf that trims the 2-player edge also lowers him at 3 and 4 players, where he sits at +0.8 and -0.7, and a 2-player-only rule is more rules weight than a 4-point gap deserves. It stays, flagged for playtesting.
- **Old Quill, 2 players: -3.8 points** (-0.5 with 3, +1.3 with 4). His ability is the weakest of the six in 2-player duels against an ability-less delver (72.8%, against 75.2%–80.8% for the others). Probably because Queens and Kings need rivals nearby to hit, and a 2-player board has only one. Strengthening him for 2 players would push him up at 4, where he is already at +1.3, so this is also left for playtesting: human players may well use his free Kings better than the AI does.
- Every other delver and player count is within 2.4 points of a fair share.

WHERE THE AI FALLS SHORT

The AIs look one turn ahead and value a hand by simple rules. They don't bluff, don't coordinate against a leader, don't plan multi-turn routes around the Warden, and don't read opponents' hands. Real players will probably gang up on the leader more than the AI does (which would help the game) and find ability tricks the AI misses: Pip & Pell's chains and Old Quill's free Kings are the most likely to play better in human hands than they do here. The simulator also can't measure fun, teachability or how long people really think. Treat these numbers as a floor for balance and a first estimate for time, then playtest with people.

REPRODUCE IT

```
cd sim && ./run_sweeps.sh v1 v2 v3 v4 reruns every sweep (about an hour per version on four cores; seeds are fixed, so the numbers come back the same). python print/designers_notes.py rebuilds these notes and CHANGELOG.md from the results.
```